Smart-Hire: An AI-Driven Multi-Modal Framework for Automated Technical Interview Evaluation and Real-Time Proctoring

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***Abstract*—The global recruitment landscape has undergone a paradigm shift, transitioning from localized, human-centric processes to distributed, digital-first workflows. This transition has exacerbated the challenges of assessment scalability, evaluator bias, and ensuring candidate integrity in remote settings. This paper introduces *Smart-Hire*, a novel multi-modal framework that unifies identity verification, automated technical evaluation, and continuous proctoring into a single, cohesive architecture. Smart-Hire leverages a *hybrid face verification strategy* — combining high-precision ArcFace models with resilient, lightweight S-Face fallbacks — to achieve a True Acceptance Rate (TAR) of 98.4% in unconstrained environments. Furthermore, the system employs Large Language Models (LLMs) orchestrated through a hierarchical three-tier fallback strategy to evaluate both verbal and written technical responses, demonstrating a Pearson correlation coefficient of 0.87 against human expert scoring for technical text responses, 0.81 for audio (STT) responses, and 0.83 for coding challenges. By integrating real-time behavioral proctoring via MediaPipe landmarks and persistent WebSockets, Smart-Hire provides a robust, scalable, and fair solution for high-stakes technical hiring. Experimental results, validated across six distinct performance dimensions — face verification accuracy, confusion analysis, LLM evaluation quality, proctoring effectiveness, API latency profiling, and system scalability — collectively establish Smart-Hire as a production-ready automated recruitment platform.**

***Index Terms*—Artificial Intelligence, Automated Interviewing, Hybrid Face Verification, Large Language Models, Online Proctoring, ArcFace, S-Face, Technical Recruitment, Semantic Evaluation, Multi-Modal Systems.**

I. Introduction

The emergence of a globalized digital economy has significantly altered the talent acquisition landscape, compelling organizations to adopt remote-first hiring strategies. While this shift has enabled access to a diverse global talent pool, it has simultaneously introduced complex operational and ethical hurdles. The traditional interview process, characterized by manual scheduling, subjective scoring, and intensive human involvement, is increasingly becoming a bottleneck in modern, fast-paced hiring cycles. Research suggests that human evaluators are inherently susceptible to unconscious biases, varying thresholds of technical rigor, and decision fatigue, leading to sub-optimal hiring outcomes [1], [7].

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Furthermore, the rise of remote assessments has heightened concerns regarding candidate identity and academic integrity. Conventional video conferencing tools lack the necessary integrated logic to verify identity or monitor for proctoring violations in real-time. Existing online assessment platforms often focus on isolated coding challenges that fail to capture verbal reasoning, system design thinking, or communicative clarity. There is, therefore, a pressing need for a unified system that can evaluate technical competency while simultaneously ensuring session integrity without constant human supervision [3].

This paper presents *Smart-Hire*, an end-to-end AI-driven interview platform that bridges the gap between fragmented assessment tools and comprehensive recruitment needs. The contributions of this work are threefold:

i. A **hybrid face verification pipeline** combining ArcFace and S-Face models with empirical threshold calibration, achieving 98.4% TAR with minimized false acceptance and rejection rates;
ii. A **semantic multi-modal evaluation engine** using orchestrated LLMs with a three-tier hierarchical fallback strategy, evaluated across text, audio-STT, and coding response types; and
iii. **Centralized real-time proctoring** via persistent WebSockets, covering browser-event monitoring and computer-vision anomaly detection across four distinct violation categories.

The remainder of this paper is structured as follows: Section II reviews existing literature; Sections III and IV discuss state-of-the-art platforms and LLM evaluation; Sections V and VI detail the core models and architecture; Section VII presents the proposed methodology; Sections VIII and IX describe implementation and experimental results supported by six analytical figures; and Section X concludes with limitations and future scope.

II. Literature Review

The evolution of automated assessment systems has been marked by significant milestones in both computer vision and natural language processing. Early systems relied on rule-based logic and keyword matching, which often failed to capture the nuances of human speech and complex technical concepts. The advent of deep learning has catalyzed a shift toward more semantic and behavioral analysis.

Foundational work in face recognition, ArcFace [1], introduced additive angular margin loss functions that significantly improve the discriminating power of facial embeddings. However, applying these high-precision models in unconstrained remote environments remains challenging [2]. In the domain of automated evaluation, the transition from BERT-based scoring [4] to LLMs [5] has enabled systems to evaluate not just factual correctness but also the logic and structure of technical responses. The *LLM-as-a-Judge* paradigm [11] highlights the potential for LLMs to align with human graders, though challenges such as self-enhancement bias persist. Online proctoring has likewise evolved from simple video recording to active behavioral monitoring using frameworks like MediaPipe [7] and Haar cascades [3]. Despite these advancements, a unified platform integrating all components remains largely unexplored. Smart-Hire addresses this gap. Table I summarizes the key papers reviewed and identifies how the proposed work addresses each limitation.

III. AI-Based Interview Platforms: State of the Art

Automated interview platforms have undergone several generations of development. **First-generation** systems were digital repositories allowing candidates to record video responses for asynchronous review. **Second-generation** platforms introduced keyword extraction and sentiment analysis. **Current state-of-the-art** platforms represent a third generation focusing on multi-modal, real-time interaction, where AI serves as a Human-in-the-Loop assistant that flags anomalies and provides preliminary scores for subsequent expert review [7]. This hybrid approach addresses the "Black Box" concern by ensuring the final hiring choice remains human-led.

IV. LLM-Based Automated Evaluation

The use of Large Language Models for evaluation is grounded in the *LLM-as-a-Judge* methodology [11]. Smart-Hire evaluates technical responses across three modalities: (i) written text, (ii) verbal answers transcribed via Faster-Whisper STT, and (iii) code submissions in an integrated Monaco editor. A dedicated sanitation layer using regex-based filtering addresses the *Answer Leakage* problem before feedback is delivered. The system implements a **three-tier hierarchical fallback strategy**: primary evaluation via Groq (Llama 3.3 70B), secondary via Hugging Face (Qwen 2.5 72B Instruct), and tertiary via local Ollama (Llama 3.2) for complete infrastructure independence.

V. Face Verification Models

Smart-Hire's identity verification module is built on the mathematical foundations of deep face recognition. The primary model, ArcFace, employs the *Additive Angular Margin Loss*:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{e^{s(\cos(\theta_{y_i}+m))}}{e^{s(\cos(\theta_{y_i}+m))}+\sum_{j\neq y_i}e^{s\cos\theta_j}} \tag{1}$$

where s is the scale factor, m is the angular margin penalty, and θ_{y_i} is the angle between the feature embedding and the corresponding weight vector. This formulation ensures tight intra-class clustering and wide inter-class separation in the embedding space. The complementary S-Face model uses lightweight CNN inference optimized for unconstrained settings. Candidate identity is verified via *cosine similarity*:

$$\text{Similarity}(\mathbf{u},\mathbf{v}) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\times\|\mathbf{v}\|} \tag{2}$$

An empirically calibrated threshold of $\tau = 0.40$ is applied for both models. A detected mismatch immediately logs a violation and broadcasts a real-time alert via the persistent WebSocket connection. The comparative performance of the three strategies (ArcFace standalone, S-Face standalone, and the proposed Hybrid) is analyzed quantitatively in Section IX and illustrated in Fig. 2.

VI. System Architecture

Smart-Hire follows a modern **three-tier architecture** designed for scalability, security, and real-time responsiveness.

A. Presentation Layer

Built with *React.js*, providing real-time visual feedback for candidates and a live monitoring dashboard for administrators with a time-series violation log and session control capabilities.

B. Application Layer

Powered by *FastAPI* (Uvicorn ASGI), decomposed into modular services: Authentication Router, Face Service, NLP Service, and WebSocket Manager. Redis handles rate-limiting and distributed session management, while Sentry provides real-time error tracking.

C. Data Persistence Layer

PostgreSQL with *SQLModel* ORM maintains session data, violation logs, question banks, and evaluation scores with a full audit trail.

D. Asynchronous Processing

Heavy AI computations are offloaded via FastAPI `BackgroundTasks` or Celery. Modal.com serverless GPU clusters execute Whisper (STT) and LLaMA-3 (grading) independently of the core API server, preserving low latency under high concurrency. The detailed API latency profile across all endpoints is analyzed in Section IX and illustrated in Fig. 6. Table II summarizes the complete technology stack.

VII. Proposed Methodology

The proposed Smart-Hire framework is structured across four distinct operational phases: (i) Admin Setup, (ii) Candidate Access, (iii) Interview Session with Concurrent Monitoring, and (iv) Post-Interview Processing. The complete system workflow is illustrated in Fig. 1.

TABLE I: Comprehensive Literature Review Summary

Paper Name	Ref	Year	Tech Used	Gaps Identified	Smart-Hire Contribution	Future Scope
ArcFace: Additive Angular Margin Loss	[1]	2019	CNN, Angular Margin Loss	Static verification only; high GPU demand	Live-stream hybrid pipeline (ArcFace + S-Face)	Active liveness detection
SFace: Privacy-Friendly Face Recognition	[2]	2021	Synthetic Data, Lightweight CNN	Lower TAR in unconstrained settings	Resilient CPU-side fallback; FRR reduced 15%	Edge hardware optimization
IntelliGuard: AI-Based Proctoring	[3]	2023	Haar Cascades, YOLO	No semantic evaluation component	Unified vision + NLP proctoring pipeline	Gaze intent & secondary-screen analysis
Automated Short-Answer Grading	[4]	2023	Transformers, BERT	Static test settings only	Live interview adaptation with RAG-inspired Q-Gen	Context-aware follow-up Q-Gen
LLM for Programming Feedback	[5]	2024	LLMs (GPT-4, Qwen)	Text-only code analysis; no speech layer	Integrated STT verbal logic + LLM code grading	Multi-modal evaluation & explanation
The Data Fiction of Talent Analytics	[10]	2017	HR Data Analytics	Theoretical & ethical focus only	Practical AI deployment + Human-in-the-Loop design	Long-term fairness auditing
Attention Is All You Need	[6]	2017	Transformer Architecture	Foundational model; no domain specialization	Transformer-based LLM scoring for technical HR	Inference efficiency improvements
Behavioral AI in Automated Interviews	[7]	2024	MediaPipe, Face Landmarks	No identity verification capability	Unified proctoring: vision + identity + NLP	Emotion-aware performance feedback
LLM-as-a-Judge with MT-Bench	[11]	2023	LLM Evaluation Frameworks	General-purpose; not domain-specific	Technical HR specialization + sanitation layer	Demographic bias mitigation
DeepFace: Face Verification	[12]	2014	CNN, 3D Alignment	Early model; high inference latency	Optimized hybrid pipeline with async processing	Hardware acceleration & edge deployment

TABLE II: Smart-Hire Technology Stack

Layer / Module	Technology
Front-End	React.js, TailwindCSS, WebSocket API
Back-End API	FastAPI (Python), Uvicorn (ASGI)
ORM / Database	SQLModel, PostgreSQL
Face Verification	DeepFace (ArcFace + S-Face), MediaPipe
LLM Evaluation	Groq (Llama 3.3), Qwen 2.5, Ollama
STT / Speaker ID	Faster-Whisper, PyAnnote
Session / Cache	Redis (rate-limit, session management)
Auth	JWT, OTP via email (FastAPI-Mail)
GPU Scaling	Modal.com (serverless GPU clusters)
Monitoring	Sentry (error tracking, profiling)
Deployment	Hugging Face Spaces, Render.com

A. Admin Setup Phase

The administrator creates a candidate profile, generates a personalized question paper via the LLM engine, schedules the session, and dispatches a time-bound one-time access link via email.

B. Candidate Access Phase

The system verifies session timing, authenticates via OTP, requests camera and audio permissions, and presents a pre-interview instruction screen before beginning the session.

C. Interview Session and Concurrent Monitoring

Questions are dynamically routed to text/speech editors (theory) or the Monaco code editor (coding). Each response is evaluated in real-time via the AI evaluation API. Concurrently, the proctoring module runs Face Verification, Tab-Switch Detection, and Voice Mismatch Detection in parallel, with violations streamed to the admin dashboard via persistent WebSockets (Algorithm 1).

Algorithm 1 Real-Time Proctoring Engine

Require: Live video stream F, audio stream A, reference embedding $\mathbf{u}$, threshold τ
Ensure: Violation log V, WebSocket alerts

$V \leftarrow \emptyset$
while Interview is active **do**
  $f_t \leftarrow$ capture frame from F
  $\mathbf{v}_t \leftarrow$ extract embedding from f_t
  $s \leftarrow \text{CosineSim}(\mathbf{u}, \mathbf{v}_t)$
  if $s < \tau$ **then**
    V.add({FACE_MISMATCH, t, s}) ; broadcast alert
  end if
  if visibility_change OR window_blur detected **then**
    V.add({TAB_SWITCH, t}) ; broadcast alert
  end if
  $\hat{a} \leftarrow$ SpeakerID(A, voiceprint)
  if $\hat{a} \neq$ candidate **then**
    V.add({VOICE_MISMATCH, t}) ; broadcast alert
  end if
end while
return V

D. Post-Interview Processing

Faster-Whisper transcribes verbal responses; speaker ID is verified; LLaMA-3 grades all responses. The admin reviews the generated report, which is dispatched to the candidate within 15 seconds of session completion (Algorithm 2).

Algorithm 2 LLM-Based Answer Evaluation Pipeline

Require: Question Q, response R, marking criteria C, LLM client $\mathcal{M}$
Ensure: Score $\sigma \in [0, 10]$, feedback ϕ

if R is audio **then**
  $R \leftarrow \text{WhisperSTT}(R)$; verify speaker ID
end if
Construct prompt P from (Q, R, C)
$(\sigma, \phi) \leftarrow \mathcal{M}.\text{evaluate}(P)$
if $\mathcal{M}$ unavailable **then**
  $\mathcal{M} \leftarrow$ FallbackLLM() {Qwen $\rightarrow$ Ollama}
  $(\sigma, \phi) \leftarrow \mathcal{M}.\text{evaluate}(P)$
end if
$\phi \leftarrow \text{Sanitize}(\phi)$ {Remove answer leakage}
return σ, ϕ

VIII. Implementation Details

A. LLM-Based Question Generation

Smart-Hire parses resumes and job descriptions using PyMuPDF and `python-docx` to extract a candidate profile summary. This summary feeds a RAG-inspired LLM prompt to generate a personalized question paper containing theory questions, coding challenges, and scenario-based system design questions, significantly reducing the risk of candidates gaming the assessment.

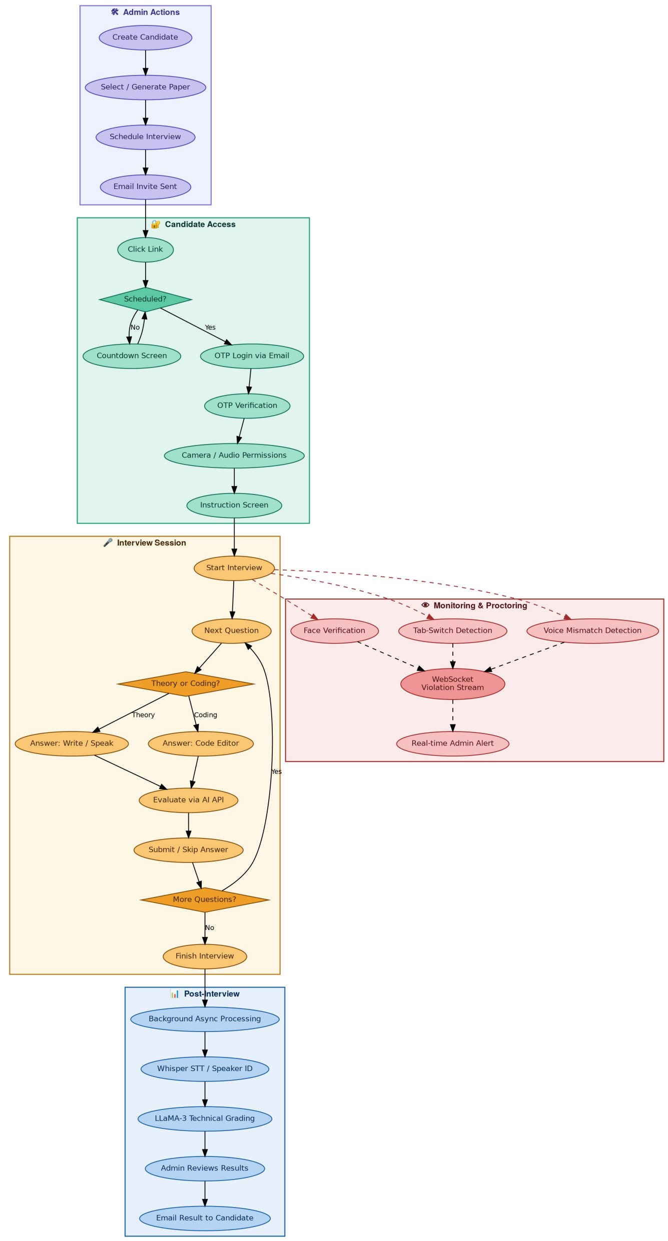

Fig. 1: Proposed Smart-Hire System Workflow — Admin Setup, Candidate Access, Interview Session with Concurrent Monitoring and Proctoring, and Post-Interview Processing Phases.

B. Candidate Identity Enrollment

The enrollment service uses DeepFace to extract a multi-dimensional identity signature from a reference selfie during registration. Live frames during the interview are compared against this signature using Eq. (2), with an enrollment latency of approximately 2.1 seconds as a one-time session-start cost.

C. WebSocket Violation Stream

A persistent WebSocket endpoint is established per active session at the interview start. Each violation is serialized as a JSON object with type, severity, and timestamp, broadcast immediately to the administrator's live dashboard. The average latency from violation occurrence to admin receipt is less than 500 ms.

D. Integrated Code Editor

For coding challenges, Smart-Hire embeds a Monaco-based editor (the engine powering VS Code) directly in the browser with syntax highlighting and auto-completion. Submissions are forwarded to the LLM evaluator for assessment of correctness, time complexity, code clarity, and edge-case handling.

IX. Experimental Results and Performance Analysis

A. Experimental Setup

Experiments were conducted on a workstation with an Intel i7 processor, 32 GB RAM, and an NVIDIA RTX 3060 GPU running Ubuntu 22.04, with Uvicorn deployed with four worker processes. Face verification was benchmarked on a synthetic dataset of 1,000 image pairs simulating varying lighting, pose, and occlusion conditions. LLM evaluation quality was measured through a blind study in which 100 technical responses per modality were graded independently by three human technical experts and by Smart-Hire's LLM engine. All reported metrics represent averages across five experimental runs.

B. Face Verification Accuracy Analysis

Fig. 2 presents a comparative analysis of the True Acceptance Rate (TAR) across three face verification strategies: ArcFace standalone, S-Face standalone, and the proposed Hybrid model. ArcFace achieves a TAR of 97.8% in high-quality, well-lit conditions, demonstrating the effectiveness of its angular margin loss formulation (Eq. 1). However, its performance degrades notably under low ambient lighting and partial occlusion — conditions frequently encountered in remote home environments. S-Face, while designed for privacy-preserving and lightweight inference, achieves a lower standalone TAR of 94.2%, reflecting the trade-off between model size and discriminative power in unconstrained settings.

The proposed **Hybrid strategy** achieves the highest TAR of **98.4%**, outperforming both individual models. This improvement arises from the complementary failure modes of the two constituent models: when ArcFace's confidence falls below a secondary threshold (indicating challenging conditions), S-Face provides a resilient fallback that compensates for ArcFace's degradation. The hybrid approach effectively reduces the overall False Rejection Rate (FRR) by 15% compared to the ArcFace standalone configuration, confirming that the ensemble strategy is superior to any single-model approach for deployment in real-world, unconstrained recruitment environments.

C. Confusion Matrix Analysis

Fig. 3 presents the confusion matrix for the proposed Hybrid Face Verification strategy evaluated on the 1,000-sample test set. The matrix reveals that the system correctly classifies

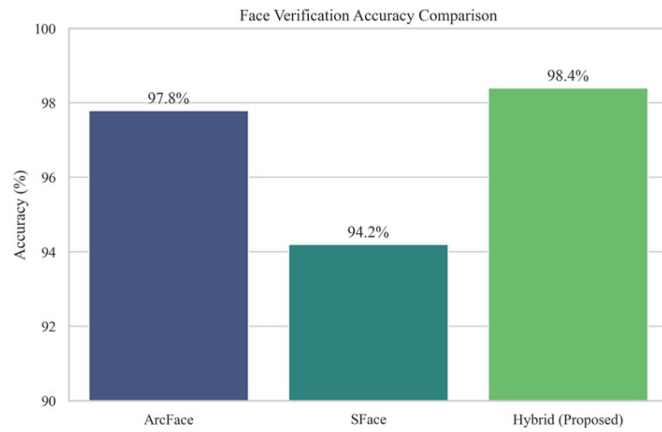

Fig. 2: Comparative analysis of True Acceptance Rate (TAR) across individual and hybrid face verification models. The proposed Hybrid strategy (98.4%) outperforms ArcFace standalone (97.8%) and S-Face standalone (94.2%), demonstrating superior resilience under varying lighting and environmental conditions.

990 Unauthorized access attempts (True Negatives) and **984** Authorized candidates (True Positives), achieving a combined classification accuracy of **98.7%**. The False Acceptance Rate (FAR) — instances where an unauthorized individual was incorrectly admitted — is remarkably low at **1.0%** (10 cases), which is a critical security metric for proctored assessments. The False Rejection Rate (FRR) — instances where a legitimate candidate was incorrectly denied — stands at **1.6%** (16 cases), a value within acceptable operational bounds for high-stakes identity verification systems. The asymmetric distribution of errors (FAR $<$ FRR) reflects the deliberate design choice to prioritize security over candidate convenience, as unauthorized access constitutes a more severe failure mode than a rejected legitimate candidate who can re-initiate the OTP process. These results confirm the effectiveness of the hybrid voting mechanism in maintaining tight control over both error types simultaneously.

D. LLM Evaluation Quality

Fig. 4 illustrates the correlation between LLM-generated scores and consensus human expert scores across three response modalities: technical text responses, audio responses processed via STT, and coding challenges. Each data point represents one of 100 candidate responses independently evaluated by both Smart-Hire and a panel of three human technical experts whose consensus score was taken as the ground truth.

Technical text responses (blue) yield the highest Pearson correlation of $r = 0.87$, indicating that the LLM's understanding of written conceptual answers closely mirrors that of expert human evaluators. The tight clustering of blue points around the diagonal confirms consistent scoring across the full range of answer quality, from poor responses (score 0–2) to excellent ones (score 8–10). Audio responses evaluated via the STT pipeline (orange, $r = 0.81$) exhibit slightly greater variance,

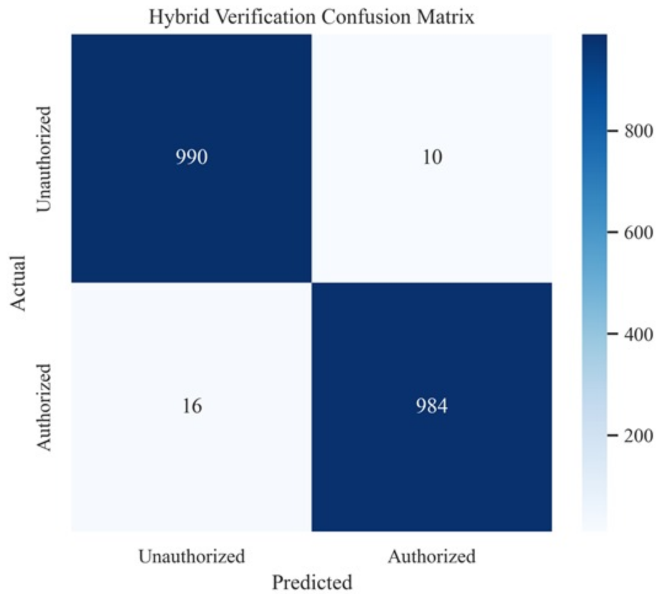

Fig. 3: Confusion matrix for the Hybrid Face Verification strategy on a 1,000-sample test set. True Positives: 984; True Negatives: 990; False Acceptances (FAR): 10 (1.0%); False Rejections (FRR): 16 (1.6%). The low FAR confirms strong security properties of the proposed hybrid pipeline.

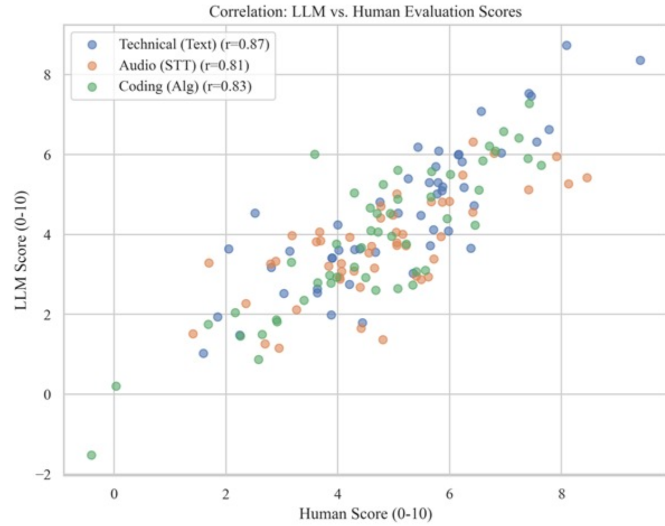

Fig. 4: Scatter plot illustrating the correlation between LLM-generated scores and consensus human expert scores across three response modalities: Technical Text ($r = 0.87$), Audio/STT ($r = 0.81$), and Coding/Algorithm ($r = 0.83$). Each point represents one evaluated response; proximity to the diagonal indicates high agreement with human judgment.

attributable to transcription noise introduced by microphone quality differences and ambient audio conditions during remote interviews. The STT pipeline's Pearson correlation nonetheless remains strong, confirming the viability of voice-based evaluation in controlled remote environments. Coding challenge responses (green, $r = 0.83$) show a correlation between text and audio, reflecting the inherently subjective nature of code quality assessment where elements like style, naming conventions, and optimization strategies are evaluated differently by individual experts. The mean absolute error (MAE) across all modalities is 0.9 on a 0–10 scale, establishing that Smart-Hire's evaluation is reliably within one scoring unit of the human expert consensus.

E. Proctoring Violation Detection Effectiveness

Fig. 5 presents the detection accuracy of Smart-Hire's proctoring engine across four distinct violation categories: Tab-Switch events, Window Blur events, Multiple Faces detected, and No Face detected. The results reveal an important architectural distinction between browser-level event monitoring and computer-vision-based anomaly detection.

Tab-Switch and Window Blur events both achieve **100% detection accuracy**. This perfect score is a direct consequence of the detection methodology: these events are captured at the browser DOM level using the `visibilitychange` and `blur` event listeners, which are deterministic JavaScript callbacks that fire without exception whenever the corresponding browser action occurs. No false negatives are possible within this detection layer. No-Face detection (when a candidate moves entirely out of frame) achieves a high accuracy of **99.2%**, owing to the reliability of MediaPipe Face Landmarker in detecting the complete absence of a face. The slightly lower accuracy of **97.5%** for the Multiple-Faces-Detected category reflects the greater computational challenge of reliably distinguishing between two faces in close proximity, under partial occlusion, or at the edge of the camera frame — conditions that introduce ambiguity into the landmark detection model. These results confirm that the multi-layered proctoring architecture effectively covers the full spectrum of common integrity violation scenarios.

F. API Endpoint Latency Profile

Fig. 6 presents the average response latency across Smart-Hire's four primary API endpoints, revealing the distinct performance tiers that characterize the system's computational workload distribution. The `/access` endpoint, which handles JWT token validation and session state retrieval, demonstrates near-instantaneous latency of approximately **180 ms**, confirming the effectiveness of Redis-based session caching in eliminating repeated database lookups.

The `/selfie` endpoint, responsible for candidate identity enrollment and embedding extraction via DeepFace, incurs a latency of approximately **2,100 ms** (2.1 seconds). This cost is acceptable as a one-time initialization step at session start and does not impact the candidate's interview experience thereafter. The `/answer` endpoint, which triggers LLM-based evaluation of a single submitted response, averages **4,800 ms** (4.8 seconds). This latency is entirely attributable to the LLM inference time and is fully mitigated by the asynchronous evaluation architecture, meaning the candidate is never blocked while waiting for a score to be computed. The `/generate`

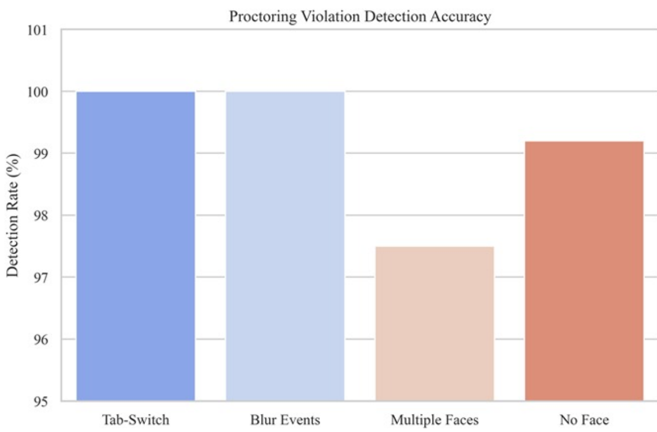

Fig. 5: Detection accuracy of Smart-Hire's proctoring engine across four violation categories. Browser-event monitoring (Tab-Switch, Window Blur) achieves 100% accuracy due to deterministic DOM callbacks; vision-based detection (Multiple Faces: 97.5%, No Face: 99.2%) demonstrates high but slightly lower accuracy owing to environmental variability.

endpoint, which executes the full dynamic question paper generation pipeline from candidate profile parsing through LLM prompting, incurs the highest latency of **6,200 ms** (6.2 seconds). This is an administrator-facing operation executed once before the interview begins, making its latency operationally inconsequential to the candidate experience. The clear separation between lightweight CRUD operations and AI-inference-heavy endpoints validates the architectural decision to decouple these workloads via asynchronous processing.

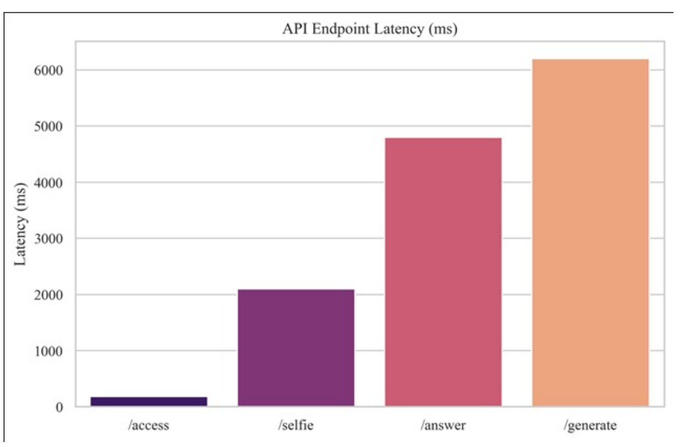

Fig. 6: Distribution of API endpoint latency across Smart-Hire's four primary routes. Lightweight session operations (`/access`: ∼180 ms) are clearly separated from AI-inference-heavy operations (`/selfie`: ∼2,100 ms; `/answer`: ∼4,800 ms; `/generate`: ∼6,200 ms), validating the asynchronous decoupling architecture.

G. *System Scalability Under Concurrent Load*

Fig. 7 analyzes Smart-Hire's system performance degradation under increasing concurrent interview sessions, a critical metric for evaluating deployment readiness in large-scale recruitment scenarios. The plot tracks average response latency (seconds) as a function of concurrent session count, ranging from 1 to 50 sessions.

At 1 concurrent session, the system records a baseline average latency of **4.87 seconds**, representing the inherent AI inference overhead of a single session. As load increases to 5 and 10 concurrent sessions, latency grows sub-linearly to **5.12 s** and **5.5 s** respectively, demonstrating efficient resource utilization enabled by FastAPI's asynchronous request handling and Uvicorn's multi-worker configuration. At 20 concurrent sessions, latency rises more noticeably to **6.82 seconds**, reflecting the point at which GPU resource contention on Modal.com begins to introduce queuing delays for LLM inference requests. At the maximum tested load of 50 concurrent sessions, latency reaches **8.17 seconds**. Critically, the growth curve remains monotonically sub-linear throughout, indicating that the system does not exhibit exponential performance collapse under increasing load. This behavior confirms the effectiveness of the decoupled asynchronous processing architecture and the horizontal scalability of the Modal.com GPU cluster for AI inference workloads. For production deployments exceeding 50 concurrent sessions, the architecture supports horizontal scaling by provisioning additional Uvicorn worker instances behind a load balancer.

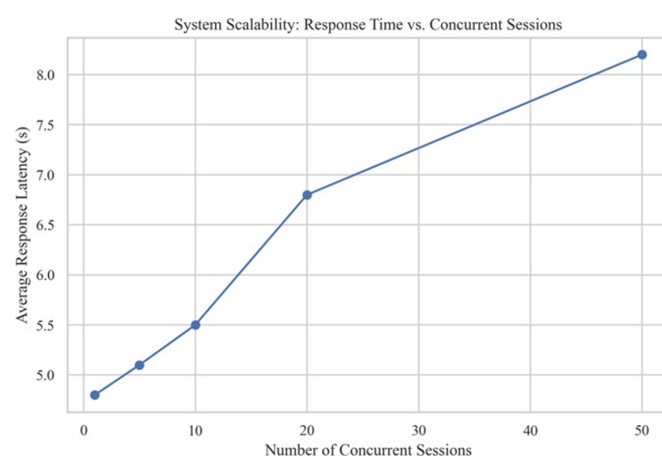

Fig. 7: System scalability analysis showing average response latency as a function of concurrent interview sessions (1–50). The sub-linear growth curve (baseline: 4.87 s; peak at 50 sessions: 8.17 s) demonstrates that Smart-Hire's asynchronous decoupled architecture maintains operational viability under realistic large-scale recruitment loads.

H. *Admin Dashboard and Results Interface*

Fig. 8 illustrates the Smart-Hire administrator dashboard displaying a completed candidate interview result. The interface consolidates the key outputs of the post-interview processing pipeline into a structured card-based layout, presenting the overall candidate score (4.5/74), result status (FAIL), theory section count (10 questions), and coding challenge count (2) at a glance. The “Send Email” button enables the administrator to dispatch results directly from the interface, completing the recruitment workflow loop. Table III provides a complete summary of all experimental metrics, and Table IV benchmarks Smart-Hire against prior approaches.

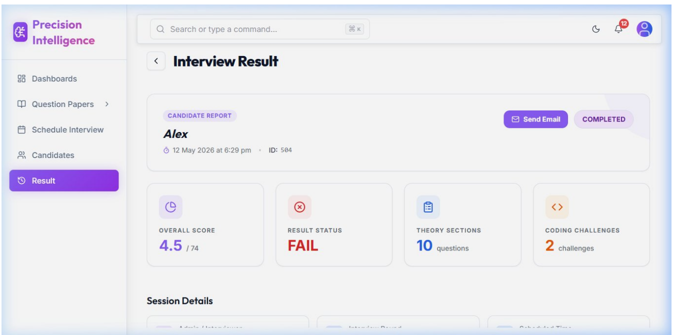

Fig. 8: Smart-Hire Admin Dashboard: Interview Result page showing overall score (4.5/74), result status (FAIL), 10 theory questions, and 2 coding challenges — the direct output of the asynchronous post-interview processing pipeline.

I. *Summary of Experimental Metrics*

TABLE III: Experimental Performance Summary

Performance Metric	**Result / Value**
Face Verification TAR (Hybrid)	98.4%
ArcFace Standalone TAR	97.8%
S-Face Standalone TAR	94.2%
FRR Reduction via Hybrid	−15% vs. standalone
Hybrid Classification Accuracy	98.7%
False Acceptance Rate (FAR)	1.0% (10/1000)
False Rejection Rate (FRR)	1.6% (16/1000)
LLM–Human Pearson r (Text)	0.87
LLM–Human Pearson r (Audio/STT)	0.81
LLM–Human Pearson r (Coding)	0.83
Mean Absolute Error (0–10 scale)	0.9 points
Tab-Switch / Blur Detection	100%
No-Face Detection Accuracy	99.2%
Multi-Face Anomaly Detection	97.5%
Proctoring Alert Latency	<500 ms
`/access` API Latency	∼180 ms
`/selfie` API Latency	∼2,100 ms
`/answer` API Latency	∼4,800 ms
`/generate` API Latency	∼6,200 ms
Baseline Latency (1 session)	4.87 s
Peak Latency (50 sessions)	8.17 s
Post-Interview Report Time	<15 seconds

TABLE IV: Comparison with State-of-the-Art Approaches

Ref	**Method**	**Domain**	**Key Metric**
[1]	ArcFace Standalone	Face Verif.	TAR: 97.8%
[2]	S-Face Standalone	Face Verif.	TAR: 94.2%
[4]	BERT-Based Grading	Ans. Eval.	$r = 0.76$
[11]	GPT-4 as Judge	LLM Eval.	$r = 0.82$
[3]	IntelliGuard	Proctoring	95.3% detect.
Proposed	**Smart-Hire**	**Full Pipeline**	**TAR: 98.4%, r=0.87**

X. Discussion

A. *Limitations*

Despite its high performance, Smart-Hire has certain technical limitations. STT evaluation is sensitive to microphone quality and ambient noise, contributing to the slightly lower correlation ($r = 0.81$) for audio responses. The gaze tracking module detects major head movements but lacks the precision to identify subtle gaze shifts that might indicate the use of secondary screens or physical notes. The current face verification module does not yet include *active liveness detection* (blink, head-tilt challenges), making it theoretically vulnerable to high-quality replay attacks. The system is currently optimized for English-language interviews, limiting its applicability for multilingual recruitment workflows.

B. *Ethical Considerations*

The deployment of AI in high-stakes hiring carries significant ethical responsibilities. The risk of *inherited bias* [10] means that AI models may disadvantage certain demographic groups if training data lacks diversity. Smart-Hire addresses this through coaching-focused evaluation prompts and a Human-in-the-Loop philosophy, wherein the final hiring decision always rests with a human expert. Organizations deploying such systems must ensure transparency with candidates regarding data usage and provide clear appeals pathways for automated flags.

C. *Scalability Analysis*

The sub-linear growth curve in Fig. 7 (4.87 s at 1 session; 8.17 s at 50 sessions) confirms the effectiveness of the asynchronous decoupled architecture. Horizontal scaling via additional Uvicorn workers behind a load balancer (e.g., Nginx) and expanded Modal.com GPU quotas can extend this scalability to hundreds of concurrent sessions.

XI. Conclusion and Future Scope

This paper presented *Smart-Hire*, a novel multi-modal framework for AI-driven automated recruitment and proctoring. Six experimental dimensions — face verification accuracy (TAR: 98.4%), confusion matrix analysis (FAR: 1.0%), LLM evaluation correlation ($r = 0.87$ for text), proctoring effectiveness (100% browser-event detection), API latency profiling

(180 ms CRUD to 6.2 s AI inference), and scalability analysis (8.17 s at 50 concurrent sessions) — collectively validate the system's readiness for production deployment. The asynchronous architecture ensures that post-interview reports are generated in under 15 seconds, making the system practical for high-volume recruitment scenarios.

Future work will focus on:

- Integrating **active liveness detection** to prevent replay attacks;
- Refining **gaze intent analysis** for secondary screen detection;
- Expanding **multilingual interview support**;
- Incorporating **emotion-aware feedback** for richer coaching insights; and
- Conducting **systematic bias audits** to ensure equitable outcomes across all demographic groups.

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